

RNA-Seq Literature Familiarization and Data Characterization

Preparation for RNA-Seq Re-Analysis using Galaxy and Github

Group Topic

Group 3 - Infection

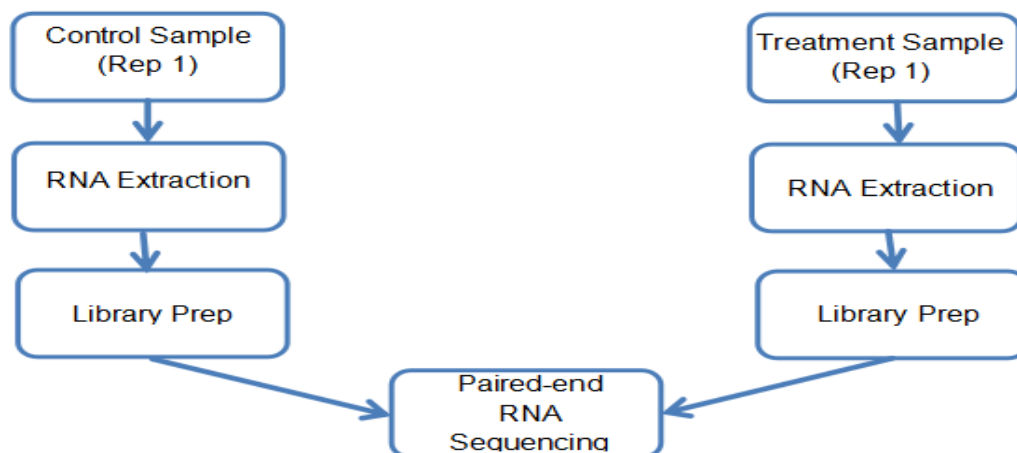
Group Members

Bahian, Celine
Duhaylongsod, Jil
Melon, Kris Bernadette
Puyat, Doreen Marie G.
Torres, Rica Jane

Information from the paper	Group Answer
Title of the Paper	RNA-Seq Analysis of the Host Response to <i>Staphylococcus aureus</i> Skin and Soft Tissue Infection in a Mouse Model.
Authors	Brady RA, Bruno VM, Burns DL
Organism	Mouse (BALB/c, 10-week-old female)
Year	2015
Journal	PLOS ONE
Tissue/organ/cell type	Ear skin/tissue
Biological question	What are the local and systemic host transcriptional responses to <i>Staphylococcus aureus</i> skin and soft tissue infection (SSTI)?
Control condition	Uninfected ear from the same challenged mouse; for the systemic response, ears from naive mice were used as the control.
Treatment or stress condition	Epicutaneous infection with <i>Staphylococcus aureus</i> USA300 strain SAP149.
No. of Biological replicates	3 naive and 3 infected mice per time point (Days 1,4, and 7)
Sequencing platform	Illumina HiSeq
Single-end or pair-end sequencing	Paired-end
Read length, if reported	100 nt from each end (100 bp paired-end)

RNA-seq repository	NCBI Sequence Read Archive (SRA)
Bio project/study accession	SRP040121
RNA accession No.	GSE56227
Reference genome or Transcriptome used by the authors	UCSC BALB/c mouse reference genome
Main RNA-seq software or tools reported by the authors	TopHat (alignment), DESeq/Bioconductor (differential expression), Ingenuity Pathway Analysis (pathway-level interpretation)

Part 4. Prepare a Simple RNA-Seq Experimental Design Diagram



What was changed or manipulated?

- Mice were exposed to *Staphylococcus aureus* infection on one ear.

What was the control?

-The non-infected ear of the challenged mouse served as the local control. The study also included ears from naïve/uninfected mice for systemic-response comparisons.

What was measured?

- Gene expression/transcriptome changes in response to the infection using RNA-seq.

How many biological replicates were used?

- 3 mice per condition/time point. The GEO dataset lists three mice for each relevant group at days 1, 4, and 7.

What tissue or cell type was sequenced?

- Mouse ear tissue (skin/soft tissue). RNA was extracted from infected and non-infected ears.

Part 5. Locate the Public RNA-Seq Data

#	Condition	Rep	SRX	Spots	Bases	Pair length (bases/spot)	Per-mate read length
1	TSTAP Day 0 (Mouse 1)	1	SPX489216	22.2M	4.5G	202 bp/spot	101 bp
13	TSTAP_Day1UninfectedMouse1	1	SRX490269	26.5M	5.4G	202 bp/spot	101 bp
29	TSTAP_Day1Mouse2Left	1	SRX490728	21.5M	4.35 G	202 bp/spot	101 bp
33	TSTAP Day 1 (Mouse 2 – Right)	1	SRX490755	17.4 million	3.51 G	202 bp/spot	101 bp
45	TSTAP_Day4UninfectedMouse1	1	SRX490858	23.6M	4.8G	202 bp/spot	101 bp

Part 6. Assign One RNA-Seq Sample to Each Student

Run accession	Condition	Bio. replicate	Single/ Paired-end	Approx. file size
SRR1192261	Controlled	1	Paired-end	2.0 GB
SRR1193316	Uninfected Mouse1-Day 1	1	Paired-end	3.6 GB

SRR1193865	Infected (local response / treatment)	Mouse 2 (of 3 infected mice at Day 1)	Paired-end	2.93 Gb
SRR1193913	TSTAP Day 1 (Mouse 2 – Right)	Rep 1	Paired-end	2.4 Gb
SRR1194016	Uninfected Mouse1- Day 4	Rep 1	Paired-end	4.31 Gb

Part 7. Download the RNA-Seq Data into Galaxy

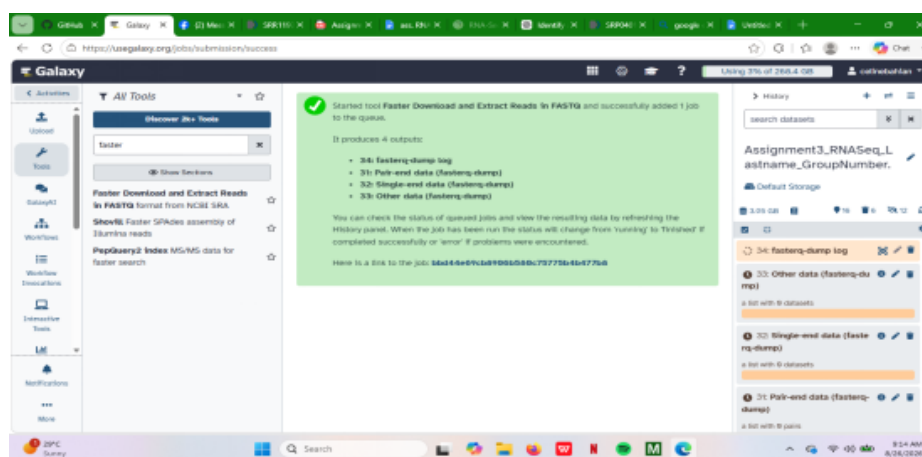


Figure 2. Successful download and extract of Pair_End and Single-End RNA Seq Reads in Galaxy using the tool FASTER Download and Extract Reads in FASTAQ.

Part 8. Examine the FASTQ File

Matrix	Control_Rep1_R1	Control_Rep1_R2
(1) Line1:sequence identifier	@SRR1192261.SN180:373:C3PLYACXX:1:1101:1406:2054/1	@SRR1192261.SN180:373:C3PLYACXX:1:1101:1406:2054/2

Line2:nucleotide sequence	CGCAGTTGGGCAGAAAGC CTCCAGAACATCACAGG GCAAGGGAGACAGCACC TGCCACGGACGTGCTAAT TCCTCAGGGCAGCCAAC CTCCGGGACAGCTGG	CGAGAGCTCTGTGCGCA GCCAGGAGGACCAGCTG TCCCGGAGGTTGGCTGC CCTGAGGAATTAGCACG TCCGTGGCAGGTGCTGT CTCCCTTGCCCTGTGA
Line3:separator line	+	+
Line4:quality-score characters	@B@FFFFFFGHHHH=EGE GHGIIJJJDGIIGCHII<=FG GHIGAGIIEHIJJJIDIHHBF CABBB;@CDDCCDACC DDDDDB?C?	@@?DFFFFFFHHHHIIJJGII HJEHGBFHDFFIGHJJJJAH IEHIHFEHFEFFACD>AC D@CCCCDDDDDBBDB9? B(:4:@43:4:C9C34?@C####
(13) Line1:sequence identifier	@SRR1193316.SN180:373:C 3PLYACXX:1:1101:1407:216 4/1	@SRR1193316.SN180:373:C 3PLYACXX:1:1101:1407:216 4/2
Line2:nucleotide sequence	CCAACTCTTGCTCCAAA AACTCTGTCGGTAGCAG CAGCTTTCAATGAAGATG AAGATAGTGAGCCAGAA GAAATGCCTCCAGAAGC AAAGATGAGGATGAA	GCTGATGTTGGTGTGTCC CTTCCAATGTTCTTCATC CTCATCTTTGCTTCTGGA GGCATTCTTCTGGCTCA CTATCTTCATCTTCATTGA AAGCTGCTGC
Line3:separator line	+	+
Line4:quality-score characters	:8=?DBDFF>FD@FF>)?DG @CBB*9?<*?***?;FDA7=;A EEEE7;BD;>A@##### #####	@@=>EC?DD;@D##### #####
(29) Line1:sequence identifier	@SRR1193865.SN180:373:C3P LYACXX:5:1101:1632:1987/1	@SRR1193865.SN180:373:C3P LYACXX:5:1101:1632:1987/2
Line2:nucleotide sequence	NGGCTTCCAGGTCTTCAGT GCTGTGACTCCCCTGGAGG AGNCTGGCCTCTAGTCTCT ATTCTCTGAATAACTTTCCC AAGGNCTCTTGATGCTCTTT CTGN	NTCTGCACCCAGCTCTCTTT GAGCCACCCCTGCTCCCA TCTGGCCAAGCAGCCGCCT CATCAAAAAGGCCCACTGC AGGCGCCACATGTCCAAAA AGGCA
Line3:separator line	+	+

Line4:quality-score characters	#1=DDFFFHGHFHIIJJJGJJI IJHIIJJJJCHIBA#0?DHCDGHI IJHGIEHHHJCHEIEDGGEICH HFEEEEFF#;,@DCCDDDD CDDDDCCC#	#1=DDFFFHHHHHJJJJJJJJJI CHIJJ:8:90BG;?)0*?9?BF(-).@ 45A'5)77;BBED8?;AB##### #####
(33) Line1:sequence identifier	@SRR1193913.SN180:373:C 3PLYACXX:5:1101:1420:199 5/1	@SRR1193913.SN180:373:C 3PLYACXX:5:1101:1420:199 5/2
Line2:nucleotide sequence	NGGAGATTGAACACTACAG TCCCAGCTGGCCCTGAA ACAATCGCTGGAAGCCT CCTTGGCAGAAACAGAA GGTCGCTACTGCGTGCA GCTCTCCCAGATTCAA	NGGAGATCTGGCTTTGA ATCTGGGAGAGCTGCAC GCAGTAGCGACCTTCTGT TTCTGCCAAGGAGGCTT CCAGCGATTGTTTCAGGG CCAGCTGGGACTGT
Line3:separator line	+	+
Line4:quality-score characters	#1=DDFFFHGHGJJJJJJJJJI IJGJIJJJJJJJJJJJJJJJIEIJJ JIIJJJJHHEHHFFCDBDD @BDDDDDDDBDDDDDD DDDDDDDDDE	#1=DDFFFHHHHGJJJJJJJJ JGJGIJJFIJJJJJJJJJJJJG HHIIJJHHHGHFFFDDEEED DDDDDDDDDDDCDCCD@ DDDBDDDD?
(45) Line1:sequence identifier	@SRR1194016.SN180:373:C 3PLYACXX:5:1101:1710:200 0/1	@SRR1194016.SN180:373:C 3PLYACXX:5:1101:1710:200 0/2
Line2:nucleotide sequence	NTCAGCTCAAGGGTGTG CGCCAGATCCCTTGCAA CAGTGGGAGCTGGTACC TATTGAAGTATTTGAGGC AAGGCAGGTGAAAGCCA GCTTCAAGAACTA	TGGCAGGGCAACCAGGC ACACAGGCTTTCAATAGT TTCTTGAAGCTGGCTTTC ACCTGCCTTGCTCAAAT ACTTCAATAGGTACCAGC TCCCACTGTTGC
Line3:separator line	+	+
Line4:quality-score characters	#4=;DDDDDFHFH?EFHFG BGBHDEHEDGGC>FEGII GIII9;AFGG=7CGHCFHC@ EGC>EEHF:A@7??AABA? B,;@>C>5?C?	BB@DF?FHHGJJJJFIFGC FHGC9=DEIGEECEHBCE :@BAA@A@ACC>CCA32 @CCDCDDD>A>3

- How is a FASTQ file different from the FASTA genome file used in the previous assignments?
 - A FASTQ file contains nucleotide sequences and their quality scores, while a FASTA file contains only the nucleotide sequences and their identifiers.

2. What information is present in FASTQ that is not present in FASTA?
 - FASTQ contains a quality-score line for each sequence. These characters indicate the confidence of each base call during sequencing.
3. Why are quality scores important in RNA-seq?
 - Quality scores indicate how reliable each nucleotide in a read is. They help researchers identify and remove low-quality reads or bases, improving the accuracy of the read alignment and gene expression analysis.

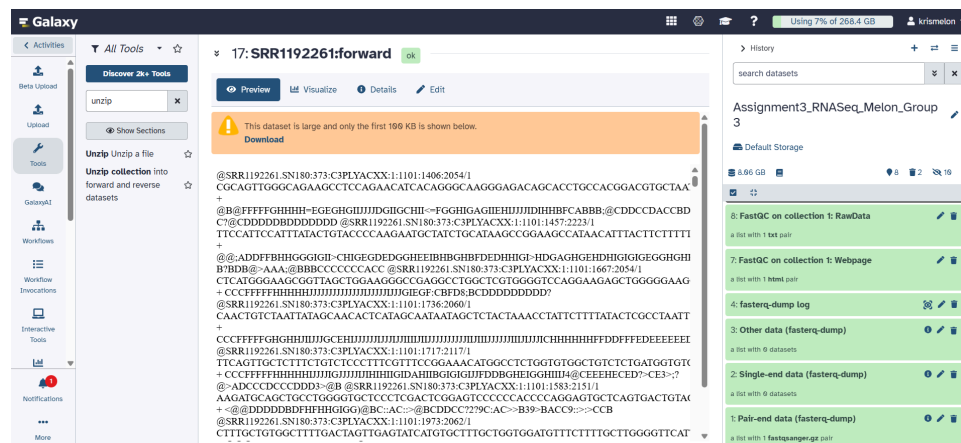


Figure 3. Preview of the forward reads of Control_Rep1_R1 from paired-end RNA sequencing data in FASTQ format, generated using the Galaxy tool Unzip Collection to Forward Datasets.

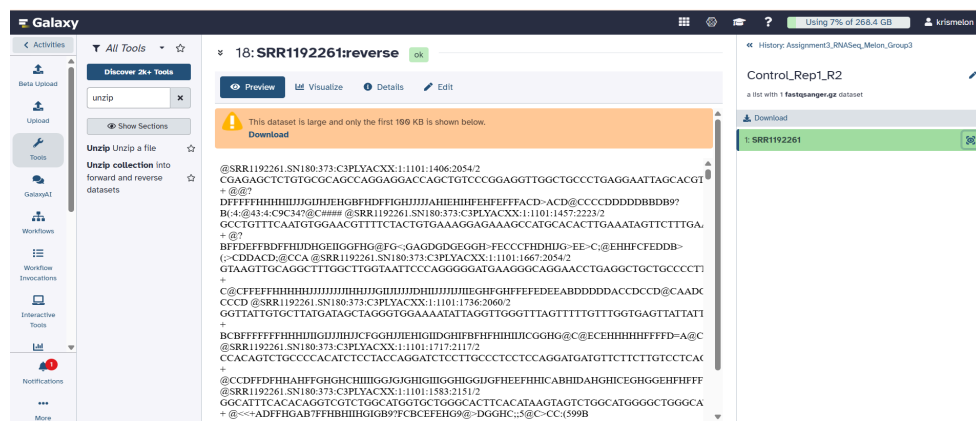


Figure 4. Preview of the reverse reads of Control_Rep1_R2 from paired-end RNA sequencing data in FASTQ format, generated using the Galaxy tool Unzip Collection to Reverse Datasets.

Part 9. Run FastQC



Figure 5. FastQC Quality Control for Forward Read (Control_Rep1_R1) generated using the FastQC tool in Galaxy.

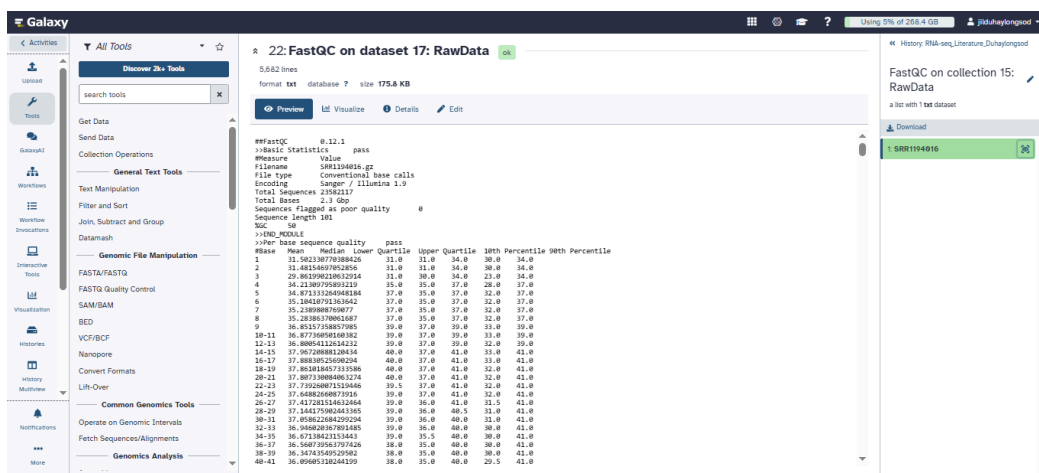


Figure 6. Raw Data Output of the FastQC Analysis for the Forward Read (Control_Rep1_R1) Generated Using the FastQC Tool in Galaxy

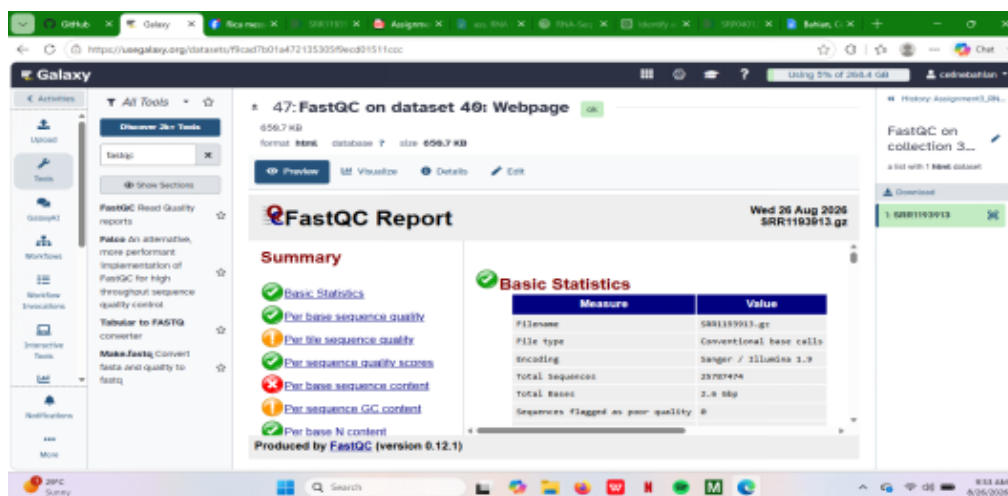


Figure 7. FastQC Quality Control Report for the Reverse Read (Control_Rep1_R2) Generated Using the FastQC Tool in Galaxy

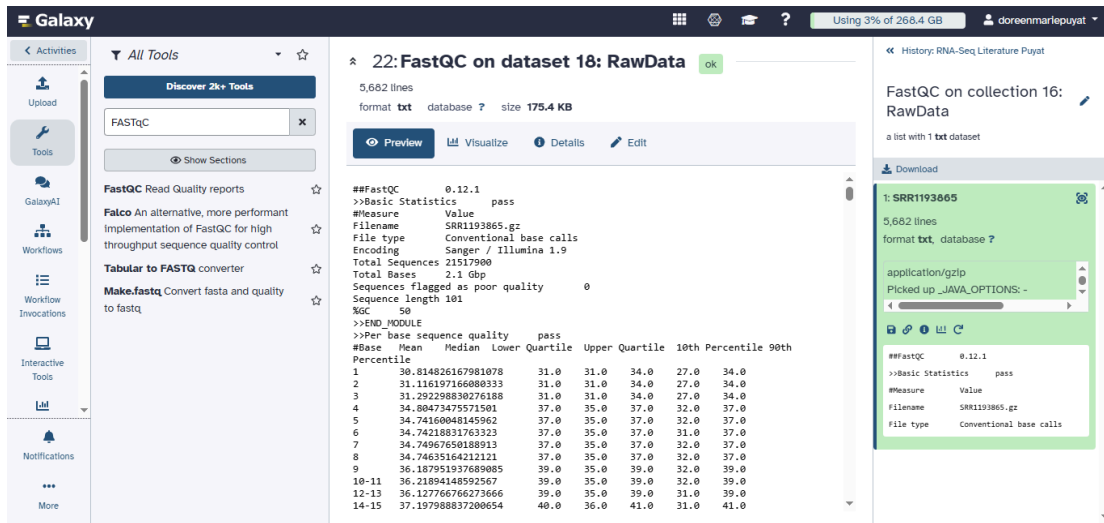


Figure 8. Raw Data Output of the FastQC Analysis for the Reverse Read (Control_Rep1_R2) Generated Using the FastQC Tool in Galaxy

Part 10. Characterize Your RNA-Seq Dataset

RNA-Seq Characteristic	Observation
(01) RNA-seq run accession	SRR1192261
Condition	Controlled - Day 0
Single-end or paired-end	Paired-end
Number of reads / sequences	22,211,000
Read length	101 bp
File size	2 GB
GC content	49.8%
General per-base sequence quality	PASS – High quality across most bases, with high per-base quality scores.
Presence of adapter content	No significant adapter contamination detected.
Presence of overrepresented sequences	No significant overrepresented sequences detected.

Any quality warning observed	No major quality warnings, overall sequencing quality is good.
(13) RNA-seq run accession	SRR1193316
Condition	Uninfected Mouse1-Day 1
Single-end or paired-end	Paired-end
Number of reads / sequences	26,527,933
Read length	101 bp
File size	3.6GB
GC content	50.1%
General per-base sequence quality	PASS – Good quality, with most bases having high Phred scores (Q30 or higher).
Presence of adapter content	No significant adapter contamination was detected.
Presence of overrepresented sequences	No major overrepresented sequences were found.
Any quality warning observed	No major quality warnings were observed, indicating good overall sequencing quality.
(29) RNA-seq run accession	SRR1193865
Condition	Infected (local response / treatment)
Single-end or paired-end	Paired-end (per the paper's Methods)
Number of reads / sequences	21,517,900
Read length	101 bp
File size	2.93 Gb
GC content	50%
General per-base sequence quality	Good. It passed for both R1 and R2, high scores overall with a normal dip near the read ends
Presence of adapter content	None detected.

Presence of overrepresented sequences	R1: minor warning; R2: none
Any quality warning observed	Yes. Warnings for per-tile quality, GC content, and duplication levels; fail on per-base sequence content (normal RNA-seq priming bias)
(33) RNA-seq run accession	R1&R2: SRX490858
Condition	R1&R2: TSTAP Day 1 (Mouse 2 – Right)
Single-end or paired-end	R1&R2: Paired-end
Number of reads / sequences	R1&R2: 25787474
Read length	R1&R2: 101 bp
File size	R1: 175.4Gb R2: 176.1 Gb
GC content	R1&R2: 50%
General per-base sequence quality	R1&R2: PASS – High-quality bases across the 101 bp reads (mostly Phred Q30–Q40)
Presence of adapter content	R1&R2:
Presence of overrepresented sequences	R1&R2:
Any quality warning observed	R1&R2:
(45) RNA-seq run accession	SRR1194016
Condition	Uninfected Mouse1- Day 4
Single-end or paired-end	paired-end
Number of reads / sequences	23,582,117
Read length	101 bp
File size	4.31GB
GC content	50.3%
General per-base sequence quality	PASS – The majority of bases have good per-base quality scores.
Presence of adapter content	No noticeable adapter contamination was detected.

Presence of overrepresented sequences	No important overrepresented sequences were detected.
Any quality warning observed	No major quality problems were observed, showing good overall sequencing quality.

Part 12. Compare the RNA-Seq Samples Within the Group

Student	Accession	Condition	Reads	Read Length	GC%	Main QC observation
Melon, Kris Bernadette	SRR1192261	Controlled-Day 0	2,211,000	202 bp	49.8%	PASS – Good overall sequencing quality with a high number of reads, 101 bp read length, and a balanced GC content of 49.8%. No major quality issues are indicated.
Torres, Rica Jane	SRR1193316	Uninfected Mouse1-Day 1	26,527,933	101 bp	50.1%	PASS – Good overall sequencing quality with a high number of reads, consistent 101 bp read length, and balanced GC content of 50.1%. No major quality issues are indicated.
Puyat, Doreen Marie G.	SRR1193865	Infected (local response / treatment),	21,517,900	101 bp	50%	Good overall quality (21.5M reads, 101 bp, 50% GC) with

		Day 1				expected RNA-seq warnings — per-tile quality, GC distribution, and duplication levels — plus a per-base sequence content fail from normal random-hexamer priming bias. No adapter contamination. R2 shows more quality drop-off toward the read end than R1.
Bahian, Celine	SRR1193913	Uninfected (local control), Day 1	25,787,474	101 bp	50%	PASS – Excellent overall quality with high per-base scores (Q30+), no poor-quality reads, and no major QC issues.
Duhaylongsod, Jil	SRR1194016	Uninfected (systemic control), Day 4	23,582,117	101 bp	50.3%	PASS – Good overall sequencing quality with a high number of reads, consistent 101 bp read length, and balanced GC content of 50.3%. No major QC issues are indicated.

Part 13. Interpretation Questions

1. What biological question was the original RNA-seq study trying to answer?

- The study wanted to determine how mouse skin and soft-tissue cells respond to *Staphylococcus aureus* infection. The researchers examined changes in gene expression over time to understand which biological processes and immune responses are activated during infection.

2. Why did the authors use RNA-seq instead of only examining the genome?

- Looking only at the genome tells us what genes are present, but not necessarily which genes are active. RNA-seq measures the RNA being produced, allowing researchers to see which genes are expressed and how their expression changes during *S. aureus* infection.

3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?

- Genomic DNA contains the organism's relatively stable genetic information, including genes that may or may not be active. RNA molecules are produced when genes are expressed, so their amounts can change depending on the cell's condition. RNA-seq mainly measures these expressed transcripts.

4. What is a biological replicate and why is it important?

- A biological replicate is an independent biological sample, such as a different mouse, that undergoes the same experimental condition. Biological replicates are important because they show whether an observed result is consistent among individuals rather than being caused by random variation in only one sample.

5. What is the difference between single-end and paired-end sequencing?

- In single-end sequencing, the sequencer reads each DNA fragment from only one end. In paired-end sequencing, the fragment is read from both ends, producing two reads from the same fragment. Paired-end sequencing can provide more information about the location and structure of sequences and can improve read alignment.

6. What is a FASTQ file?

- A FASTQ file is a file format that stores sequencing reads together with their quality scores. Each read normally consists of four lines: the sequence identifier, nucleotide sequence, separator line, and quality-score characters.

7. What information does FastQC provide?

- FastQC evaluates the quality of sequencing data before further analysis. It provides information such as per-base sequence quality, sequence quality scores, GC content, sequence length, duplicated sequences, overrepresented sequences, and possible adapter contamination.

8. What does a high per-base quality score indicate?

- A high per-base quality score indicates that the sequencer has high confidence that the nucleotide was called correctly. In general, higher Phred scores correspond to a lower probability of a sequencing error.

9. Why can adapter contamination be a problem?

- Adapters are artificial sequences added during library preparation. If they remain in the sequencing reads, they can interfere with read alignment and downstream analysis. Adapter contamination can also affect sequence-quality measurements and may produce misleading results.

10. Were all RNA-seq samples in your group similar in quality? Explain.

- Yes. Based on the FastQC results, the RNA-seq samples were generally similar in quality. The forward and reverse reads both showed high per-base sequence quality, mostly around Q30–Q40, and both passed the main quality checks.

11. Did any sample show a possible quality problem? What was it?

- No major quality problems were observed in the FastQC results. The reads had good overall quality, with no significant adapter contamination or overrepresented sequences. Therefore, the samples appeared suitable for further RNA-seq analysis.

12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

- Before comparing gene expression, the researchers would need to perform quality control and trimming, remove adapters and poor-quality bases, and then align or quantify the reads against the reference genome/transcriptome. The reads would be counted for each gene, followed by normalization to account for differences in sequencing depth. Finally, statistical analysis would be performed to identify genes that are significantly differentially expressed between the control and treatment groups.

Part 15. Group Summary

Our group selected an RNA-seq study involving mouse (*Mus musculus*) TSTAP tissue because it demonstrates how RNA sequencing is used to compare gene expression between control and treatment conditions. The experiment included 12 RNA-seq samples, consisting of six control and six treatment replicates, and was sequenced using the Illumina HiSeq 2000 platform with paired-end 2×101 bp reads. By examining the raw FASTQ files and FastQC reports, we learned how to evaluate sequencing quality through per-base quality scores, GC content, sequence length, and adapter contamination. The FastQC results showed that the samples were consistently high quality, with most bases scoring above Q30 and no major quality issues or adapter contamination. After quality control, the original authors aligned the reads to the mouse reference genome, quantified gene expression, and performed differential gene expression analysis to identify genes affected by the treatment. To reproduce part of their analysis, we will need to learn read alignment, gene counting, normalization, and differential expression analysis using RNA-seq bioinformatics tools.

Article Link

Brady, R. A., Bruno, V. M., Burns, D. L., & Subbian, S. (2015). RNA-Seq analysis of the host response to *Staphylococcus aureus* skin and soft tissue infection in a mouse model. PLOS ONE, 10(4), e0124877. <https://doi.org/10.1371/journal.pone.0124877>

Galaxy Link

Bahian: <https://usegalaxy.org/u/celinebahian/h/assignment3-rnaseq-bahian-group-3>

Melon: <https://usegalaxy.org/u/krismelon/h/assignment3-rnaseq-melon-group3>

Puyat: <https://usegalaxy.org/u/doreenmariepuyat/h/rna-seq-literature-puyat>

Torres: <https://usegalaxy.org/u/ricajanetorres/h/assignment3-rnaseq-torres-group03>

Duhaylongsod:

<https://usegalaxy.org/u/jilduhaylongsod/h/assignment3-rnaseq-duhaylongsod-group03>